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Presented at

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FCEN Facultad de Ciencias Exactas y Naturales

Universidad Nacional de Colombia

Reconstruction of Spatio-Temporal Criminal Networks for Disruption in the Amazon

From partial communications to unobserved vertices and edges: mathematical construction, statistical evidence, and applied decision-making

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FastAnalytics S.A.S · Scientific presentation — mathematical audience version

2nd International Workshop on Statistics 2026

Universidad Nacional de Colombia, Manizales Campus · Methods, Modeling and Applications

The study area: the Amazon shared by Colombia, Venezuela and Brazil

Fuente: current article, Data Collection: "to identify the existing network structure in a study area larger than France but slightly smaller than Turkey" — at scale: larger than Spain (502k km²) and close to Turkey (799k km²)

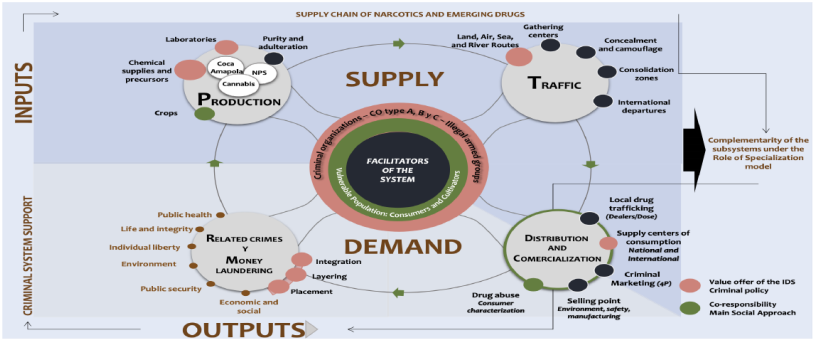




Síntesis. ≈ 706k km² of rainforest: 40% larger than Spain, 12% smaller than Turkey.

The Illicit Drug System (IDS): where the network lives

Fuente: Rodríguez (2023), translated and adapted · thesis defense, slide 1



Illicit Drug System (IDS), built and translated from [Rodríguez, 2023].

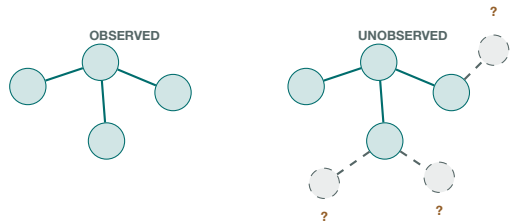
Síntesis. The IDS shows trafficking is one link of a system; our network is its logistical layer.

Intelligence observes a subnetwork, not the full network

Fuente: current article, Introduction and Related Work, pp. 2–5

Interceptions reveal **locations and sequences**, but not all actors nor all connections.

The problem: it is not only to measure the observed network, but to infer which structure is missing *without turning a prediction into a fact*.



Síntesis. Object of inference: unobserved elements conditioned on partial evidence and territorial context.

RESEARCH QUESTION

The question is stated on unobserved vertices and edges

Fuente: current article, Introduction, pp. 3–4

Can we reconstruct a spatio-temporal criminal network by predicting its unobserved vertices and edges?

$G = (V, E)$ with $E \subseteq V \times V$ (directed: order matters)

Vertices: cells with communications · Edges: ordered pairs (u, v) of consecutive activity

1 · Build
Observed chronnet

→

2 · Reconstruct
Supervised prediction

→

3 · Validate
Operational intelligence

Known
V_K, E_K – directly intercepted

Unknown
candidates: pairs (u, v) without evidence

Hidden
V_H – beyond geographic reach

Síntesis. The unit of analysis shifts from isolated events to a network with known, unknown and hidden elements.

586 communications define the 2018 observable window

Fuente: current article, Data Collection, pp. 6–8

586	6	304	2018
HF records	groups A–F	active days	Jan 2 – Dec 31

OBSERVATION LIMITATIONS

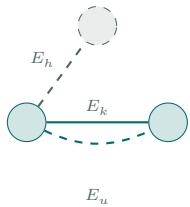
approximate location · multiple speakers · mobile transmitters · overlapping territories · dual affiliations · unmonitored areas.

Síntesis. Every record contains time and location; its absence does not prove absence of activity.

The typology separates the observed from what will be inferred

Fuente: current article, Table 1, p. 9

Element	Notation	Definition
Known vertex	V_k	Location observed in intelligence
Known edge	E_k	Temporal link between known vertices
Unknown edge	E_u	Predicted between known vertices
Hidden edge	E_h	Involves at least one hidden vertex
Hidden vertex	V_h	Location inferred through hidden edges
Unknown vertex	V_u	Beyond the study's geographic reach

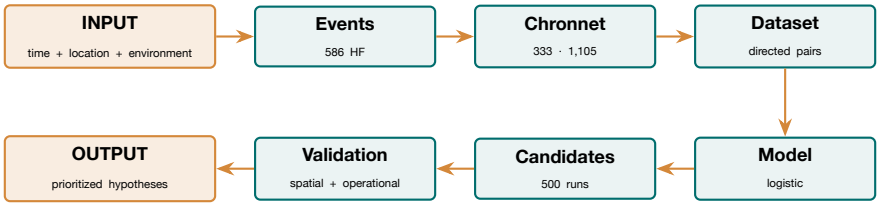


solid: known
dashed teal: unknown
gray dotted: hidden

Síntesis. A hidden edge involves at least one hidden vertex; not necessarily two.

The method chains construction, learning and validation

Fuente: current article, Analytic Approach and Results, pp. 9–21



Síntesis. Each stage produces a distinct object: network, dataset, classifier, candidates and validation evidence.

The basic datum is an event ordered in space and time

Fuente: current article, Chronological Criminal Network, pp. 9–10

$$e_a = (l_a, t_a), \qquad l_a = (x_a, y_a)$$
$$t_1 \leq t_2 \leq \cdots \leq t_{N_E}$$



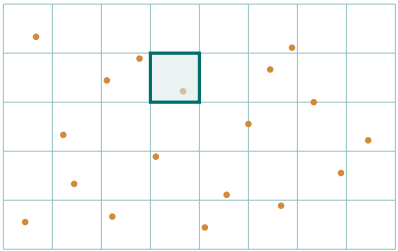
time orders events before connecting them

Each communication provides an **approximate location** and a **timestamp**. Chronological order precedes any connection.

Síntesis. Events are ordered first; then aggregated into cells and windows.

Spatial resolution controls event aggregation

Fuente: current article, Table 2, p. 10



the **cell** (0.03°) defines the vertex

≈ 3.33 km · area ≈ 9.6 km^2

Cell	Min.	Q1	Med.	Mean	Q3	Max.
0.01°	1	1	1	1.24	1	12
0.02°	1	1	1	1.48	1	22
0.03°	1	1	1	1.76	2	25
0.04°	1	1	1	2.00	2	25
0.05°	1	1	1	2.20	2	39
0.06°	1	1	1	2.47	2	43

Chosen: 0.03° ≈ 3.33 km · area ≈ 9.6 km^2

Síntesis. The cell defines the vertex: coarser resolution aggregates more events but reduces spatial precision.

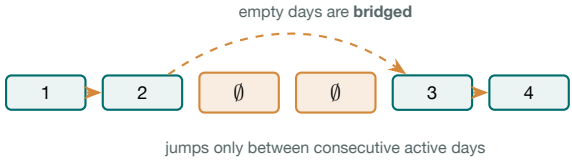
Connections occur between consecutive active days

Fuente: current article, Temporal Aggregation and Windows, p. 12

$$I_k = [(k-1)\Delta t,\ k\Delta t)$$

$K = 304$ active days

$\Delta t = 1$ day (sensitivity: $\Delta t = 7$ days)



Only windows with **at least one event** enter.

Example.

Jan 2: 3 communications · Jan 3: none · Jan 4: none · Jan 5: 5 communications

The chronnet connects Jan 2 → Jan 5 directly: the two empty days create no vertices and do not break the sequence.

Why does it matter?

The criminal network does not stop on days without interceptions: patrols, equipment failure or operational silence. Bridging the gaps avoids artificially fragmenting the temporal chain.

Síntesis. A jump joins consecutive active days, not necessarily consecutive calendar days.

Complete connection between active cells of t_n and t_{n+1}

Fuente: current article, Analytic Approach, eq. (1), p. 13 · pedagogical numerical example

WEIGHT OF EACH ORDERED PAIR · EQUATION 1

$$w_{ij} = \sum_{k=1}^{K-1} n_i^{(k)}\, n_j^{(k+1)}$$

$n_i^{(k)}$: events of cell i on active day k.
$K = 304$: active days of 2018.
The product aggregates *co-activity*; the sum accumulates transitions.

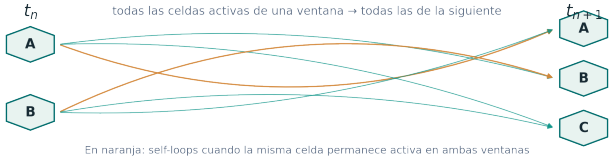
EXAMPLE · A SINGLE PAIR

active day 1: A·2 → active day 2: B·3

A → B: $2 \times 3 = 6$

contribution of pair A→B; does not imply an exclusive connection

IMAGE · COMPLETE RULE BETWEEN WINDOWS



every active origin connects to **all** active destinations; the same cell may form a self-loop (orange)

The accounting identities verify the aggregation

Fuente: current article, eqs. (2) and (3), p. 13

LIMITING CASE · ONE EVENT PER WINDOW

$$w_{ij} = \sum_{a=1}^{N_E-1} \mathbb{I}(\kappa(l_a) = i)\, \mathbb{I}(\kappa(l_{a+1}) = j)$$

$$\sum_{i,j} w_{ij} = N_E - 1 \quad \text{accounting identity}$$

GENERAL FORM

$$\sum_{i,j} w_{ij} = \sum_{k=1}^{K-1} N^{(k)} N^{(k+1)}$$

with $N^{(k)} = \sum_m n_m^{(k)}$

$N_E - 1$
total transitions

→

$\sum_k N^{(k)} N^{(k+1)}$
co-activity accumulated per window

→

verification
detects aggregation errors

Síntesis. Total weight is fixed by event counts: it detects aggregation errors.

The observed network contains 333 vertices and 1,105 edges

Fuente: current article, Tables 3 and 5, pp. 11 and 19

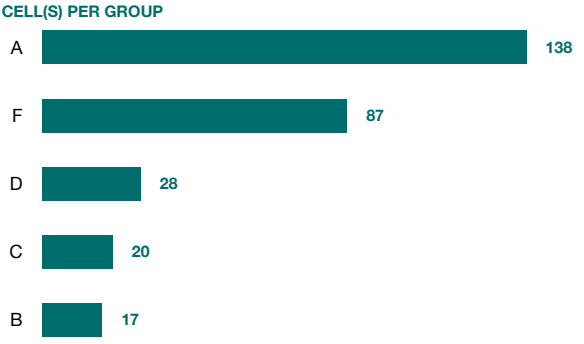
333

occupied cells (vertices)

1,105

directed edges

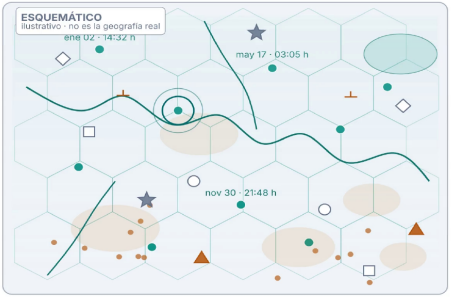
Frequencies describe presence in cells; they are not by themselves a centrality measure.



Síntesis. The observed network is the reference to evaluate reconstruction and discover missing elements.

Every vertex inherits its environmental context

Fuente: Amazonia presentation (2026), slide 7 · schematic, article layers



Capas de contexto (se acumulan)

- Comunicaciones HF interceptadas
  puntos con sello de hora
- Incendios NASA FIRMS/VIIRS
  año 2018
- Pérdida de bosque Hansen
  años 2000–2021
- Clima WorldClim + elevación
  tiempo de viaje a ciudad/puerto
- Hidrografía OSM
  ríos y cuerpos de agua
- Infraestructura OSM (validación)
  aeródromos · combustible · policía
  cultivos · canteras · muelles

Fuentes (artículo):
HF: Policía Nacional · NASA FIRMS/VIIRS
Hansen/GEE · WorldClim · OpenStreetMap

«cada vértice hereda su contexto ambiental completo»

Síntesis. The bone is the HF communications; the flesh, remote and climatic layers; the contrast, OSM infrastructure to validate proximity.

Operational data and satellite imagery

Fuente: artículo actual, Data Collection, pp. 6–8 · presentación Amazonía (2026), diapositiva 7

Dataset	Type	Source / resolution	Use in the model
Intercepted HF communications	Operational record	National Police, 2018 · 586 records with timestamp and approximate coordinates	Chronnet vertices and events; text frequency covariates per cell
Active fires	Satellite imagery	NASA FIRMS/VIIRS, year 2018 · 375 m pixels	Signal of human activity in the (burning linked to airstrips and crop
Forest loss	Satellite imagery (time series)	Hansen/GLAD via Google Earth Engine, 2000–2021 · 30 m pixels	Recent deforestation as an envi mental predictor of the vertex

Síntesis. Three raw sources: what was intercepted (HF) and what was seen from space (fires, deforestation).

Climate, precipitation and hydrography

Fuente: artículo actual, Data Collection, pp. 6–8 · presentación Amazonía (2026), diapositiva 7

Dataset	Type	Source / resolution	Use in the model
Climate and elevation	Environmental raster	WorldClim v2.1 (19 bioclimatic bands) + SRTM elevation · 1 km pixels	Environmental covariates f_u, f_v o PCA (80% variance)
Monthly precipitation	Climate series	IDEAM, 1991–2020 normals · northern Amazon	Temporal contrast of activity month lag)
Hydrography	OSM vector	OpenStreetMap / Overpass Turbo · rivers and water bodies	Distance to drainage: riverine traf ing corridor

Síntesis. The vertex's physical environment: WorldClim climate, IDEAM rainfall and the river network as a logistical corridor.

Critical infrastructure and processing pipeline

Fuente: artículo actual, Data Collection, pp. 6–8 · presentación Amazonía (2026), diapositiva 7

Dataset	Type	Source / resolution	Use in the model
Critical infrastructure	OSM vector	OpenStreetMap / Overpass Turbo · airstrips, fuel, police, crops, quarries, docks	Spatial proximity validation (5–30 buffers)

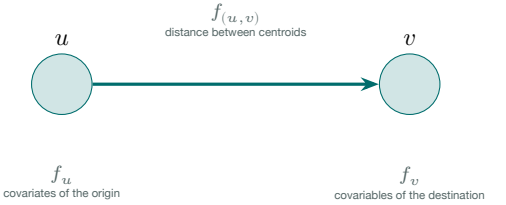
Processing pipeline

Download (FIRMS · GEE · WorldClim · Overpass) → clip to study area → extract per 0.03° cell → standardize → PCA → pair covariates.

Síntesis. Seven sources in all merge into a single vector per vertex; OSM infrastructure also validates by proximity.

Each directed pair becomes a classification observation

Fuente: current article, Criminal Chronnet Reconstruction and Data Setup, p. 14



$$P_{u,v} = \Psi(f_u, f_v, f_{(u,v)})$$

Where do the covariates come from?

Summaries of communications per cell: text length, frequency, groups present, duration; and of the pair: distance, direction and chronnet weights.

TARGET VARIABLE

$$G_{uv} \in \{0, 1\}$$

1 = observed edge
0 = pair without observed edge

self-loops excluded from this dataset

PROBLEM SCALE

333 vertices $\Rightarrow 333 \times 332 \approx 111$ thousand directed pairs
1,105 observed edges $\Rightarrow \approx 1\%$ positives

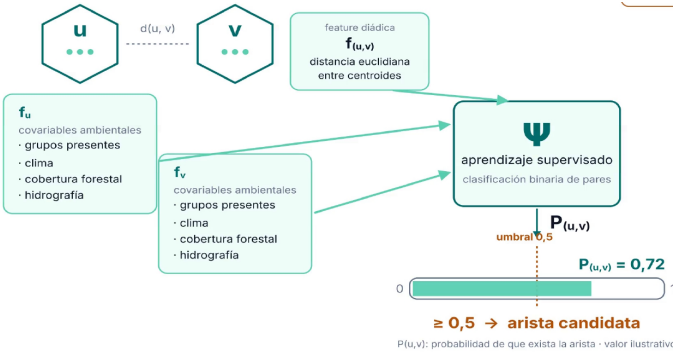
WHY NOT A TRIVIAL MODEL

Predicting “always 0” is 98% accurate and discovers nothing: the imbalance forces optimizing specificity, not accuracy.

Síntesis. Reconstruction is formulated as binary classification over pairs of cells.

Edge prediction = binary classification (schematic)

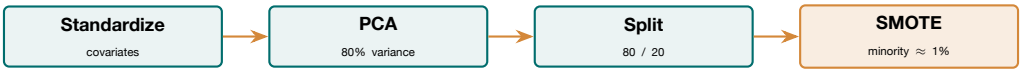
Fuente: Amazonia presentation (2026), slide 13 · schematic



Síntesis. One row per pair (u, v): covariates of both vertices plus centroid distance feed Psi; the 0.5 threshold decides candidacy.

PCA reduces dimension; imbalance redefines the problem

Fuente: current article, Data Setup and Addressing Class Imbalance, pp. 14–15



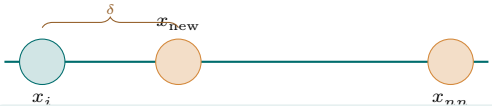
With ≈ 1% existing edges, predicting *“no edge”* for all pairs would yield a **deceptively high accuracy**.

Síntesis. The metric must reflect the cost of false positives, not only overall accuracy.

SMOTE generates variety; specificity limits candidates

Fuente: current article, Addressing Class Imbalance and Model Selection, pp. 15–17 · Chawla et al. (2002)

1 · INTERPOLATE THE MINORITY CLASS



$$x_{\text{new}} = x_i + \delta\,(x_{nn} - x_i), \quad \delta \in [0, 1]$$

$$\lambda = \left(\frac{1-r_0}{r_0}\right)\left(\frac{n_0}{n_1}\right) - 1$$

n_0, n_1: counts of non-edges and edges · r_0: target proportion · λ: required oversampling.

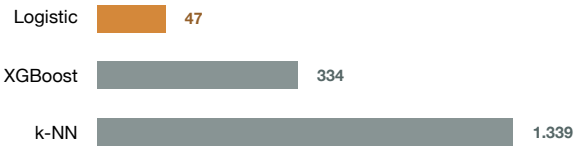
Goals: 20% evaluation · 70% candidate discovery

2 · CONTROL THE VALIDATION WORKLOAD

$$\text{Specificity} = \frac{TN}{TN + FP}$$

FP = potential candidate and verification workload; not a confirmed edge

FALSE POSITIVES · TEST



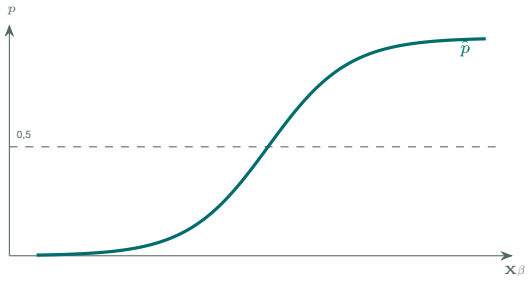
Síntesis. Fewer FP → a more manageable list for analysts; not more certainty about each candidate.

Logistic regression minimizes the false-positive list

Fuente: current article, Final Model Specification, p. 17

$$\mathbf{p} = \frac{1}{1 + e^{-\mathbf{X}\beta}}$$

$\mathbf{X}$: principal components (80% of environmental and geographic variance)



99.79%

specificity

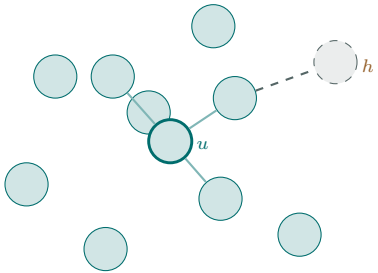
47 FP

versus 334 and 1,339

Síntesis. It is chosen for the operational criterion of specificity, not for any universal superiority of the classifier.

Homophily reduces the search space of hidden vertices

Fuente: current article, Data Processing for Unobserved Elements, pp. 17–18



$k = 3$ neighbors per vertex

hidden ones share environment (homophily)

333 known

environmental PCA

$k = 3$ **neighbors**

per known vertex

Candidate group

reduced search

500 seeds

selection stability

Assumption: hidden vertices share environmental characteristics with the observed ones. It is a *homophily hypothesis*, not a proven property.

Síntesis. KNN restricts where to search; the 500 iterations measure selection persistence.

Searching for the hidden without getting lost in the grid (flow)

Fuente: Amazonia presentation (2026), slide 18 · schematic of the real process



Diagrama esquemático del proceso descrito en el artículo (Pérez-Coronado, Canavire-Bacarreza & Lanlado) · cifras del estudio

Síntesis. Homophily → PCA of the whole grid → KNN k=3 → candidate pool → 500 iterations with distinct seeds → aggregation.

Prediction frequency measures stability across seeds

Fuente: current article, Identification of Unobserved Elements, p. 20

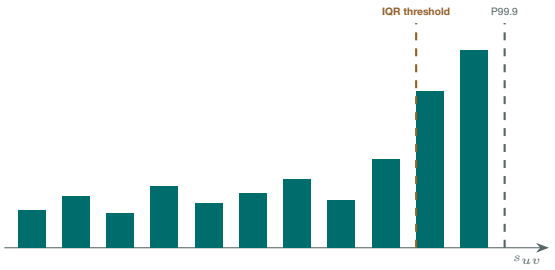
$$s_{uv} = \sum_{r=1}^{500} \mathbb{I}\left(p_{uv}^{(r)} > 0.5\right)$$

s_{uv}: positive classifications of pair (u, v) across 500 seeds (auxiliary presentation notation)

TWO CRITERIA

IQR
unknown edges

99.9th percentile
hidden edges



selection frequency over 500 runs

Síntesis. A high frequency indicates persistent selection; it is not equivalent to a probability of existence.

Reconstruction preserves the main structure of the chronnet

Fuente: current article, Table 5, p. 19

Metric	Observed	Reconstructed
Vertices	333	331
Edges	1,105	1,073
Density	0.00999	0.00982
Mean degree	6.64	6.48
Clustering	0.107	0.107
Diameter	16	16

97%

edges recovered

99.4%

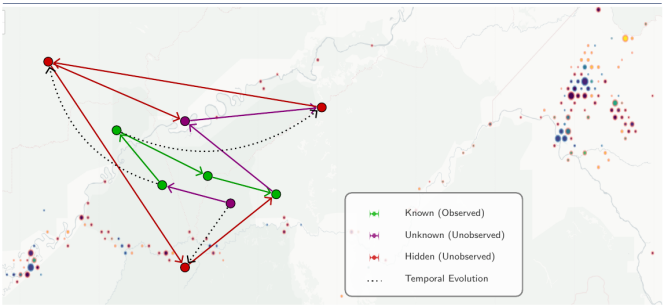
vertices retained

The comparison evaluates structural preservation; the exponents are 2.65 and 2.70.

Síntesis. Reconstruction preserves the main structure of the observed chronnet.

The reconstructed network on the territory

Fuente: thesis defense, slide 2



Legend: — known (observed) · — unknown (reconstructed) · — hidden (unobserved) · ..... temporal evolution

Síntesis. Hidden vertices (orange) close the network: they connect known circuits through unintercepted territory.

The IQR threshold selects 107 edges between known vertices

Fuente: current article, Predicted Unknown Edges, p. 20

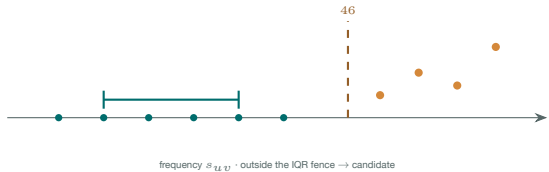
107

candidate unknown edges

$$s_{uv} > Q_3 + 1.5\,IQR = 46$$

the rule selects atypically high frequency values across the 500 runs

Both endpoints are **known vertices**.



Síntesis. The result is a prioritized list for verification, not 107 confirmed connections.

Two procedures converge on 84 unique hidden vertices

Fuente: current article, Predicted Hidden Vertices and Operational Validation, pp. 20–22

68 + 62

hidden edges: original + expanded network

COMPOSITION OF THE 84 HIDDEN VERTICES



84

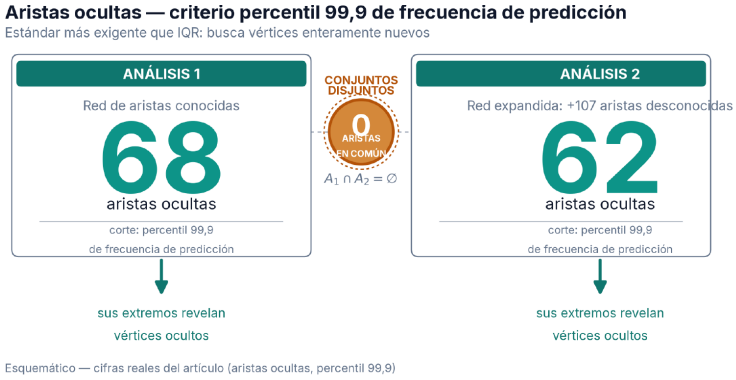
Disjoint edge sets · 19 common vertices

Validation: operational relevance confirmed by CENAC-DIJIN and DIRAN.

Síntesis. Operational validation supports relevance; it does not imply point-by-point physical verification.

Hidden edges: the standard rises to the 99.9th percentile

Fuente: Amazonia presentation (2026), slide 21 · schematic



Síntesis. 68 hidden edges in the original network and 62 in the expanded one: disjoint sets (zero in common); their endpoints reveal hidden vertices.

From a rank test to a territorial decision rule

Fuente: current article, Statistical Proximity Validation and Tables 6–9, pp. 18–25 · spatial classification, not accusation

HYPOTHESIS ON MEDIANS

$$H_0 : \text{Mediana}(C_{\text{hidden}}) \geq \text{Mediana}(C_{\text{random}})$$
$$H_1 : \text{Mediana}(C_{\text{hidden}}) < \text{Mediana}(C_{\text{random}})$$

One-tailed Wilcoxon rank test · 500 repetitions · 5, 10, 20 and 30 km buffers.

$$R = \frac{1}{500} \sum_{r=1}^{500} \mathbb{I}(p_r \leq 0.05)$$

R: proportion of significant tests (auxiliary notation)

OPERATIONAL RULE FROM THE ARTICLE



Application: quarries → attraction · docks → trade-off · always by scale and vertex group.

Síntesis. A mixed or inconclusive pattern does not imply irrelevance; it demands reading by infrastructure and scale.

The spatial strategy changes by vertex group and scale

Fuente: current article, Tables 6–9 and Discussion, pp. 21–25

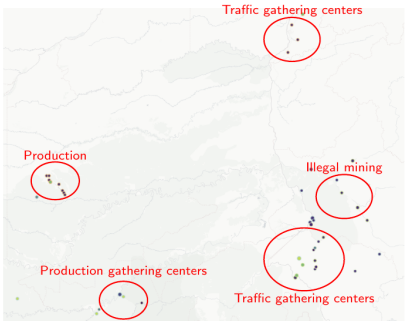
Population	Documented avoidance / attraction
333 known	Avoidance grows with scale · quarries: attraction
35 predicted	Progressive isolation · quarries: attraction
30 predicted	Avoidance · quarries and docks: attraction at 10 km
19 common	Police, quarries and docks: attraction at 5 km
19 common	Villages: avoidance at 10 km

Docks show trade-off in several populations: logistical need with interdiction risk.

Síntesis. The classification is not universal: it must be read by infrastructure, distance and vertex population.

The territory speaks: high-clustering zones

Fuente: thesis defense, slide 9



How to read it

Circles: zones with the highest local clustering coefficient. They coincide with production, gathering centers and illegal mining.

Connection with the rule

The spatial pattern validates the hypothesis: hidden vertices locate near critical infrastructure (airstrips, docks, settlements) within a 10 km buffer.

Síntesis. The coincidence between high clustering and infrastructure is spatial evidence independent of the prediction.

Chronnet definitions and identities

Fuente: current article, eqs. (1)–(3), p. 13

(1) Weight of the ordered pair

$$w_{ij} = \sum_{k=1}^{K-1} n_i^{(k)} n_j^{(k+1)}$$

(2) Limiting case

$$w_{ij} = \sum_{a=1}^{N_E-1} \mathbb{I}(\kappa(l_a) = i)\mathbb{I}(\kappa(l_{a+1}) = j)$$

(3a) Accounting identity

$$\sum_{i,j} w_{ij} = N_E - 1$$

(3b) General identity

$$\sum_{i,j} w_{ij} = \sum_{k=1}^{K-1} N^{(k)} N^{(k+1)}, \quad N^{(k)} = \sum_m n_m^{(k)}$$

Síntesis. Equation (2) is the limiting case; equation (3) is the general accounting identity.

Spatial relationships of the 333 known vertices

Fuente: current article, Table 6, p. 22

Distance	Avoidance	Trade-off	Attraction
5 km	Fuel, police	Airstrip, village, hamlet, crop, dock	Quarry
10 km	+ airstrip, crop	Hamlet, village, dock	Quarry
20 km	+ village, hamlet	Dock	Quarry
30 km	All of the above	Dock	Quarry

How to read the table

Each row accumulates ("+") the categories of smaller distances: at 30 km the network already avoids *everything*. **Avoidance** = infrastructure that cells with criminal activity tend not to have nearby; **attraction** = what they do have.

Operational data point

The quarry appears as attraction at **all** distances: material extraction (aggregate for airstrips) coexists with trafficking logistics. Statistical rules: Wilcoxon tests by infrastructure and scale (slide 23).

"+" accumulates the categories of smaller distances.

Síntesis. The quarry is the only attraction consistent at all distances for known vertices.

Spatial relationships of predicted vertices

Fuente: current article, Tables 7 and 8, pp. 23–24

Group	Distance	Highlighted result
35 · original network	5–30 km	Progressive avoidance; quarry: attraction at all scales
35 · original network	30 km	Avoids all but crops; dock/crop/hamlet: trade-off
30 · expanded network	5 km	Village: avoidance · quarry: attraction
30 · expanded network	10 km	+ airstrip/crop · quarry and dock: attraction
30 · expanded network	20–30 km	Broad avoidance · quarry: attraction

Group 35 · original network

The 35 vertices of the main subgraph. At 30 km they avoid almost all infrastructure except crops: they operate in isolated rural corridors, not in population centers.

Group 30 · expanded network

30 candidate vertices from the reconstruction. They replicate the pattern of group 35 (broad avoidance, quarry attraction) and add dock as attraction at 10 km: riverine operation.

See Tables 7–8 of the article for the full breakdown by infrastructure.

Síntesis. Both groups share growing avoidance and quarry attraction; they differ in docks and mixed categories.

Common vertices and reading controls

Fuente: current article, Table 9 and Analytic Approach, pp. 13 and 25

Distance	Avoidance	Attraction
5 km	None	Police, quarry, dock
10 km	Village	Quarry, dock
20 km	None	Dock
30 km	None	None reported

19 vertices common to both networks.

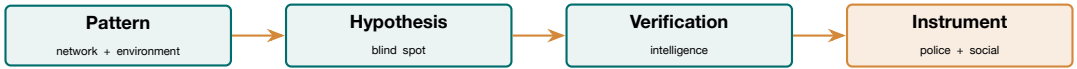
CONTROLS

accounting identities verified · $\Delta t = 7$ sensitivity · no pruning of weak edges · 500 seeds for stability · distinct thresholds for unknown and hidden edges.

Síntesis. Reproducibility requires preserving source, notation, seeds, criteria and the distinction between prediction and validation.

Evidence guides verification and complementary intervention

Fuente: current article, Conclusions and Future Work, pp. 25–27



SOCIAL CAPITAL

relations, facilitators and logistical support that sustain the operation.

RESPONSE

police action complemented with economic alternatives and institutional presence.

Síntesis. The model prioritizes questions and verification; it does not determine guilt nor prescribe automatic intervention.

Interpretation must respect the limits of the evidence

Fuente: current article, Discussion and Conclusions, pp. 21–27

OBSERVATION

Incomplete and potentially non-random data.

SELECTION

Stability thresholds, not empirical certainty.

VALIDATION

Operational confirmation as a proxy, not exhaustive verification.

SCOPE

Spatial associations, not automatic causal inference.

Síntesis. A useful prediction is a prioritized hypothesis that must undergo independent verification.

Reconstructing blind spots with interpretable evidence

Fuente: current article, Conclusions and Future Work, pp. 25–27

84	107	97%
hidden vertices identified	candidate unknown edges	edges recovered in-sample

METHODOLOGICAL EXTENSION

maritime routes, air corridors and digital networks, with data specific to each domain.

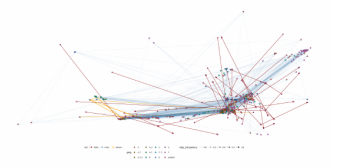
Síntesis. The applied value lies in revealing where to look and what to verify, not in automatically declaring where the crime is.

The reconstructed chronnet, in the abstract and on the map

Fuente: thesis defense, slide 12



The complete reconstructed network, visualized by connection strength.



The same complete network, visualized by its real-world latitude and longitude.

Left: the full network by connection strength · Right: the same network at its real coordinates (lat-lon).

Síntesis. Reconstruction yields a complete navigable network: the abstract view reveals hubs; the geographic one, the routes.

Thank you



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FastAnalytics S.A.S · The future speaks in data

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`fcen.unal.edu.co/2wie2026`

Workshop and materials (EN · ES): `fastanalytics.co/chronnet-taller`